

Manufacturing Predictive Maintenance Blueprint

A practical enterprise decision guide from DigiScience Techsol

ASSET TYPE: PILOT PLANNING WORKSHEET

Illustrative planning method | Updated 20 September 2026

Use this worksheet to scope one plant workflow before proposing production rollout. It contains planning fields drawn from DigiScience's manufacturing service page, not measured customer outcomes, a validated prediction model or a safety approval.

Name the bounded decision

Record one target line or asset class, the maintenance signal to examine, the accountable business owner and one measurable KPI. Describe the existing action when a signal arrives. Keep plant experts accountable for operational decisions. The first proof may address one asset class, one inspection workflow or one maintenance signal. Do not combine all plant problems into a single uncontrolled pilot.

Assemble the input inventory

- Sensor history and alarms: identify the approved source and the target asset or line.
- Maintenance work orders and failure records: identify what maintenance action and failure information are available.
- Asset hierarchy and production context: document how the signal relates to the equipment and its operation.
- Quality events, operator notes and approved visual data: include these when relevant to the agreed workflow.
- Constraints: record the data access, integration path, operating restrictions and safety requirements that affect the pilot.

Map the signal to an action

Proposed reference flow: OT or IoT sources -> secure ingestion and storage -> feature or vision pipeline -> model and evaluation -> operator workflow -> CMMS or quality action -> monitoring and audit. For each stage, record the available source/system, responsible owner, expected output and unresolved dependency. The diagram is a planning reference, not an integration already deployed at a customer.

Choose measures before the pilot

Use the measures relevant to the selected problem: unplanned downtime, MTBF, maintenance lead time, false-alert rate, defect escape rate, review time, operator acceptance, and cost per prediction or inspection. Record the current baseline, proposed measurement source, accountable owner and agreed acceptance criterion. Leave unavailable values unknown. Do not insert a promised reduction, an invented baseline or a universal pass threshold.

Worked fictional sensor and failure records

All dates, values, assets and events below are invented to explain the worksheet. They are not real plant data, physical alarm limits or evidence of a model's performance. Seven rows are insufficient to establish a reliable maintenance model.

The fictional asset is Pump-A. Each observation is at 08:00 UTC on the stated August 2026 date. Fields are: record ID; observation date; vibration RMS in mm/s; temperature in degrees C; failure within the next 24 hours. The label is 1 only when an invented confirmed failure occurs after observation and no later than 24 hours later; otherwise it is 0 only with complete follow-up. Incomplete follow-up would be Unknown, not 0.

- P01; Aug 01; 2.1; 58; label 0. Complete fictional 24-hour follow-up, no failure.
- P02; Aug 02; 2.5; 60; label 0. Complete fictional 24-hour follow-up, no failure.
- P03; Aug 03; 5.4; 78; label 1. Fictional bearing failure recorded at Aug 03 20:00 UTC.
- V01; Aug 06; 2.0; 57; label 0. Complete fictional 24-hour follow-up, no failure.
- V02; Aug 07; 5.7; 81; label 1. Fictional bearing failure recorded at Aug 07 20:00 UTC.
- T01; Aug 10; 5.2; 76; label 1. Fictional bearing failure recorded at Aug 10 20:00 UTC.
- T02; Aug 11; 2.2; 59; label 0. Complete fictional follow-up through Aug 12 08:00 UTC, no failure.

For P03, the invented failure is 12 hours after observation, so the 24-hour outcome label is 1. The later failure/work-order record can define a retrospective label; it must not be supplied as an input that was supposedly available at 08:00. A labelled event is not a model prediction or an alert that was actually delivered.

Design evaluation before fitting a model

Use earlier periods for development and later periods for evaluation. In this illustration P01-P03 form a development group, V01-V02 a validation group and T01-T02 a final holdout. The gaps leave each earlier 24-hour label window complete before the next group starts. The seven rows illustrate boundaries only; no model is fitted and no accuracy is reported.

For an actual dataset, define the prediction horizon, sensor lookback, failure definition and missing-data handling with the plant owner. Separate periods by enough time to prevent overlapping label/feature windows or shared failure events leaking across splits. Keep preprocessing and threshold choice inside development/validation data; do not tune on the final holdout. If claiming performance on new assets, reserve appropriate assets as well as future time. Record sampling intervals; an irregular event log is not automatically an equally spaced time-series dataset.

Compare the proposed alert workflow with the current process or an agreed simple baseline over the same eligible periods. Before testing, define what counts as one alert episode and one matched failure so repeated alarms cannot inflate counts. Record detected and missed failures, false alert episodes per asset operating time, alert lead time, missing inputs and operator disposition. Give denominators and uncertainty; no universal pass threshold is set by this guide.

Leave the result sheet blank until an actual evaluation: evaluation version; eligible assets/periods; failure count; detected/missed events; false alerts; exposure time; lead-time distribution; baseline comparison; reviewer decision. Record zero denominators as not evaluable, not perfect performance. A plant expert reviews any proposed action; this worksheet does not authorize automatic maintenance or a production deployment.

Technical reference for chronological splits and optional separation gaps:

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.TimeSeriesSplit.html (checked 20 September 2026). The fictional records and plant evaluation decisions here are DigiScience illustrations, not a scikit-learn maintenance standard.

Pilot evidence and controls

The pilot brief should identify the use-case baseline, approved sample data, a bounded prediction/classification/anomaly/vision/knowledge workflow, and governance controls. Plan human approval, audit trail, data classification, monitoring and failure-mode review. A scale recommendation should identify production architecture, integration backlog, cost view, rollout risks and success criteria. Feasibility evidence precedes production rollout; queue labels and dashboards do not establish that maintenance actions are staffed or effective.

Decision and source

Choose a bounded pilot only after reviewing data sources, integration, operating constraints, safety requirements and baseline measures. Scope implementation separately from a Solution Assessment; this PDF itself performs neither service. Source:

<https://digiscience.techsol.com/industries/manufacturing-ai/> Discuss one plant workflow:
<https://digiscience.techsol.com/contact>

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